Lab 3 – Architecture Selection & Justification

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Problem Statement: Hyperlocal Courier Dispatch & Tracking Engine
Architecture: Microservices Architecture

Architecture Selection

We chose Microservices Architecture for the Hyperlocal Courier Dispatch & Tracking Engine.

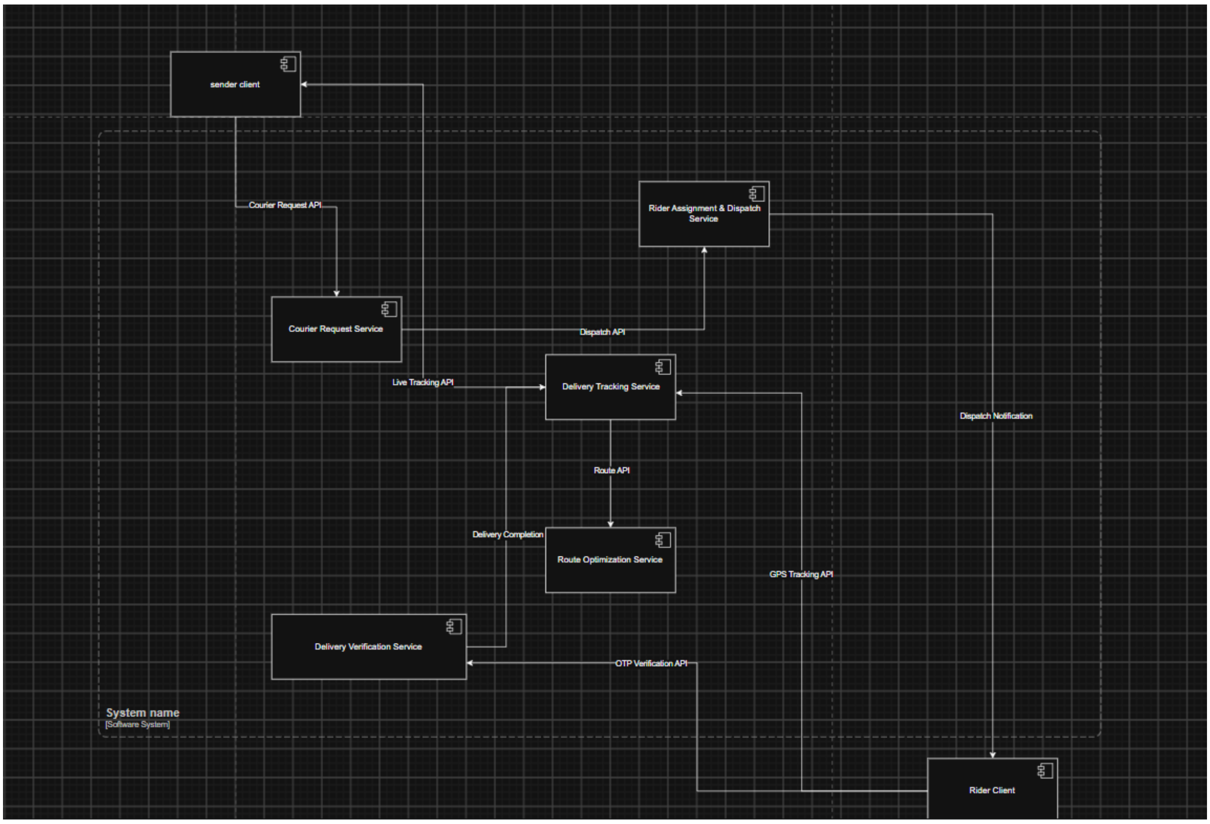
Architectural Choice: The system is divided into independently functioning services such as the Courier Request Service, Rider Assignment & Dispatch Service, Delivery Tracking Service, Route Optimization Service, and Delivery Verification Service. This structure matches the different responsibilities of the courier system.

Reason 1: Microservices allows the rider assignment and dispatch functionality to operate independently from other services. This is useful because the system must find the nearest active rider within a 3 km radius and send a dispatch notification to that rider.

Reason 2: The architecture is suitable for the system's real-time tracking and route optimization requirements. The Delivery Tracking Service can handle GPS updates independently, while the Route Optimization Service can process multi-stop routes without requiring the entire system to be modified.

Security Advantage: Separating services limits access between different parts of the system. In particular, delivery verification can be isolated as a separate service responsible for OTP verification before a delivery is marked as completed.

Performance Benefit: Individual services can handle their workloads independently. The Delivery Tracking Service can therefore focus on GPS and delivery-status updates, helping the system meet the requirement of delivering tracking updates with under 2-second latency.



sender client
Courier Request API
Rider Assignment & Dispatch Service
Courier Request Service
Dispatch API
Live Tracking API
Delivery Tracking Service
Dispatch Notification
Route API
Delivery Completion
Route Optimization Service
GPS Tracking API
Delivery Verification Service
OTP Verification API
System name
[Software System]
Rider Client